

(a) POPE responses



POPE asks, e.g.
“Is there a dog in
the image?”

Half present, half absent

Answer **Object present** Object absent

Yes

Hit

dog: “Yes”

False alarm

book: “Yes”

No

Miss

bed: “No”

Correct rejection

chair: “No”

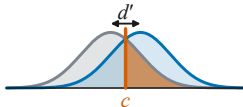
$$H = \frac{\text{hits}}{\text{present}} \quad \text{FA} = \frac{\text{false alarms}}{\text{absent}}$$

(b) Same accuracy, different responses

Discriminability $d' = z(H) - z(\text{FA})$

Criterion $c = -\frac{1}{2}[z(H) + z(\text{FA})]$

A



Object absent

Object present

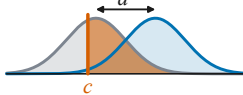
False alarms

Accuracy 0.69

H 0.69 FA 0.31

d' 1.0 c 0.00

B



Accuracy 0.69

H 0.99 FA 0.61

d' 2.0 c -1.27

Evidence that the object is present $\rightarrow$

(c) Over stages

Accuracy 0.86–0.87
at every stage

